

ADVANCED COMPUTER LAB

International Hotel Booking Analytics

Milestone 1: Report

Team 34

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Data Engineering Questions

Question 1

Which city is best for each traveler type? For each traveler type, recommend the best city based on the given reviews.

Discussion:

To answer this question, we need to access the traveler type from the users table, the city from the hotels table, and to know which hotel is "best", we want the overall score which is found in the reviews table; hence, we use the merged dataframe of the 3 tables using the user_id and hotel_id to merge. Then we want to group by traveller type and city so that we can have all Solo together for example and within the Solo we have all the cities grouped together and for each city we get all its ratings together and we specify that we want the score_overall attribute only. Next we find the mean as there might be several users of the same type who visited the same city, so we want to get the average of their scores. After that we sort with the mean score descendingly so we get the highest score first. Next we group by the traveler type and select the first to have each traveler type alone and this first corresponds to the best city for each type.

Result:

TRAVELLER TYPE	CITY	SCORE
BUSINESS	DUBAI	8.965668
COUPLE	AMSTERDAM	9.096989
FAMILY	DUBAI	9.214381
SOLO	AMSTERDAM	9.108454

Analysis:

1. Overall Findings:

The analysis reveals that **only 2 cities dominate across all 4 traveler types**: Dubai and Amsterdam. This is a striking finding that suggests these two cities excel at providing quality experiences across diverse traveler demographics.

2. City Performance:

Amsterdam (Winner for 2/4 categories)

- **Best for:** Couples and Solo travelers
- **Scores:** 9.097 (Couples), 9.108 (Solo)
- **Why it excels:**
 - High overall satisfaction across the board
 - Solo travelers give it the highest score (9.108), suggesting excellent infrastructure for independent exploration
 - Couples rate it highly (9.097), indicating romantic appeal and couples-friendly amenities
 - Known for walkability, cultural attractions, and vibrant atmosphere

Dubai (Winner for 2/4 categories)

- **Best for:** Business and Family travelers
 - **Scores:** 8.966 (Business), 9.214 (Family)
 - **Why it excels:**
 - **Families** give it the **highest score overall (9.214)** - makes sense given Dubai's family-friendly attractions (theme parks, malls, beaches)
 - Business travelers rate it highly (8.966), reflecting good business infrastructure and facilities
 - Known for luxury hotels, modern amenities, and diverse entertainment options
-

3. Score Analysis:

Score Range:

- **Highest:** Family travelers in Dubai (9.214)
- **Lowest:** Business travelers in Dubai (8.966)

- **Range:** 0.248 points (relatively narrow)

Key Observations:

1. **All scores are high (8.97-9.21)** - indicates these top cities genuinely provide excellent experiences
 2. **Family travelers are most satisfied (9.214)** - families have the highest expectations and Dubai meets them
 3. **Business travelers have the lowest score (8.966)** - though still very high, suggests business travelers may be more critical or have different priorities
 4. **Small score differences** - all traveler types are very satisfied with their recommended city
-

4. Traveler Type Insights:

Business Travelers → Dubai (8.966)

- Prefer cities with strong business infrastructure
- May prioritize: conference facilities, transportation, connectivity, professional services
- Dubai's score suggests good business hotels but room for improvement compared to leisure categories

Couples → Amsterdam (9.097)

- Seek romantic experiences, culture, and ambiance
- Amsterdam offers: canal cruises, historic charm, cozy cafes, art museums
- High score reflects romantic appeal and couples-oriented experiences

Families → Dubai (9.214) - Highest Score

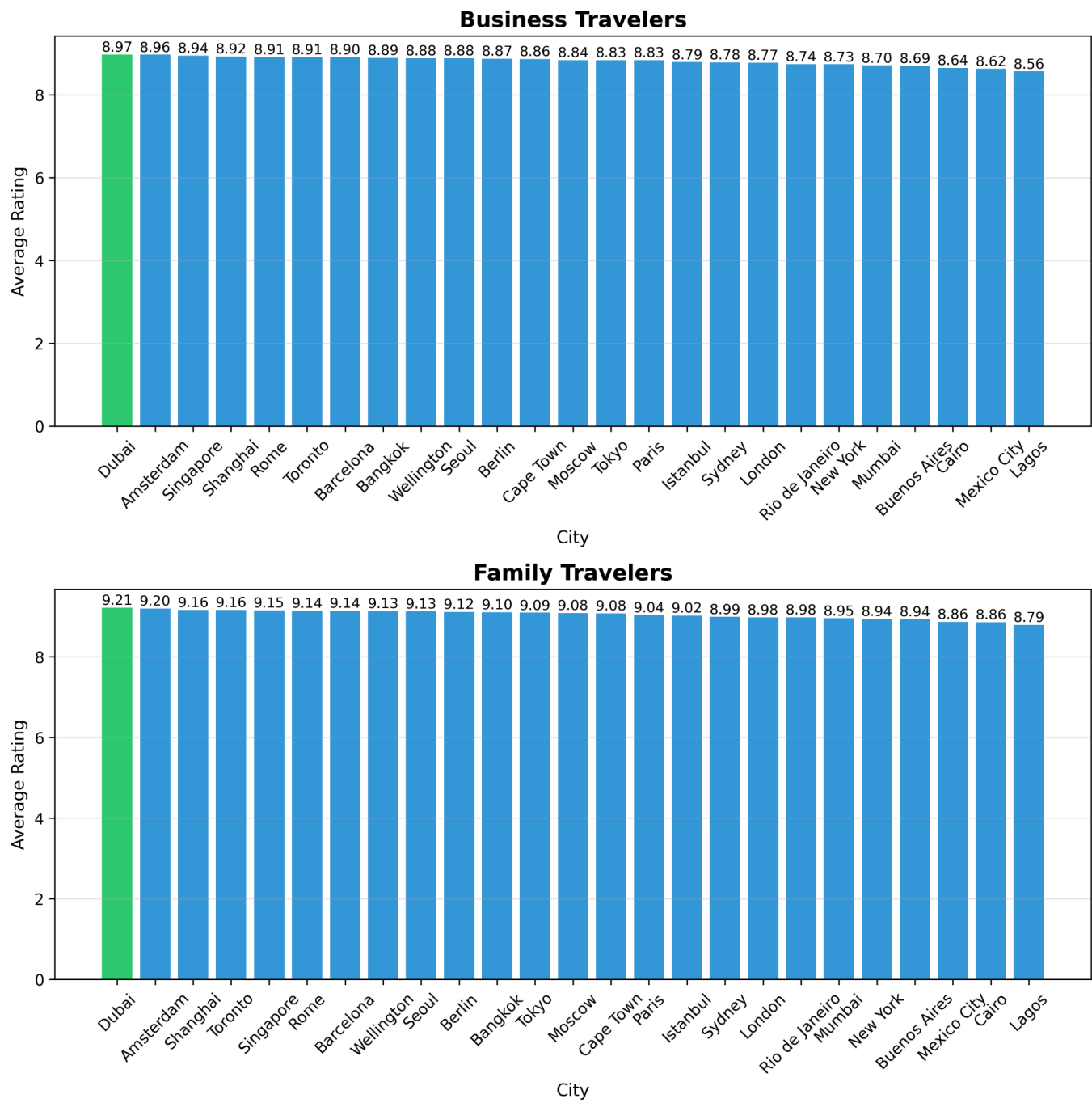
- Need child-friendly attractions, safety, convenience
- Dubai excels at: theme parks (e.g., IMG Worlds), beaches, large hotels, family activities
- Highest score indicates Dubai is exceptionally well-suited for families

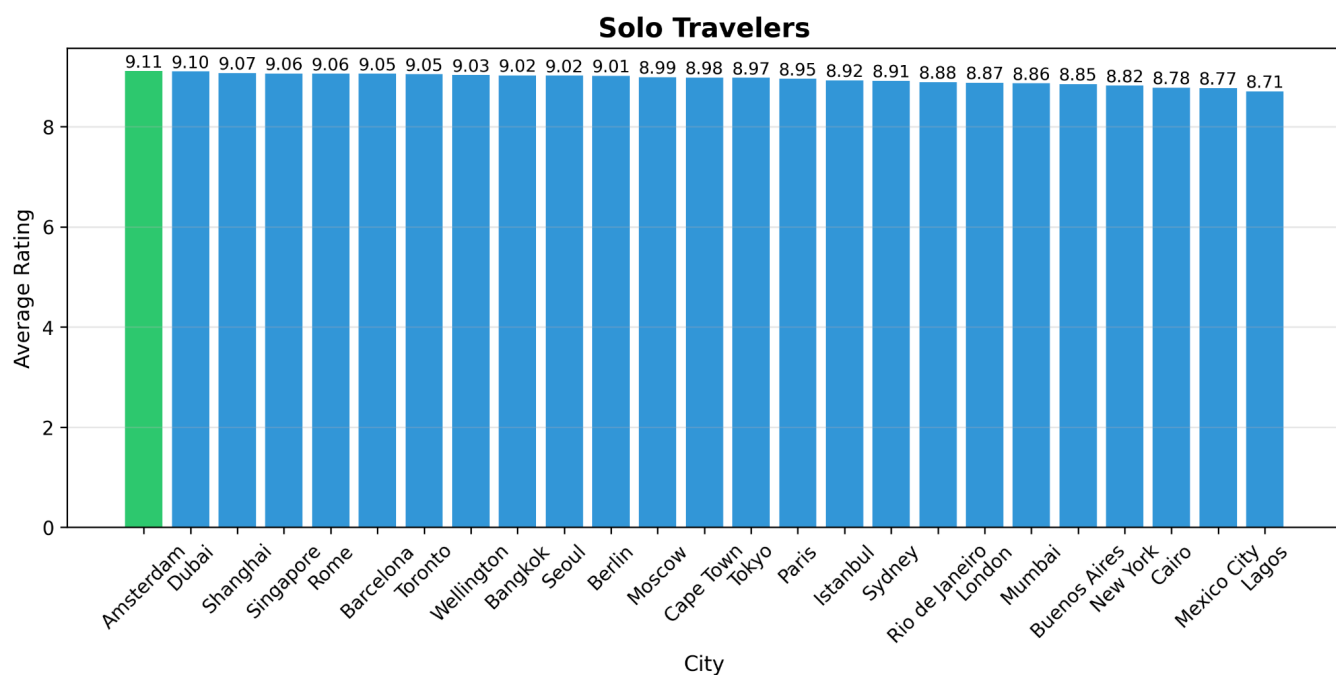
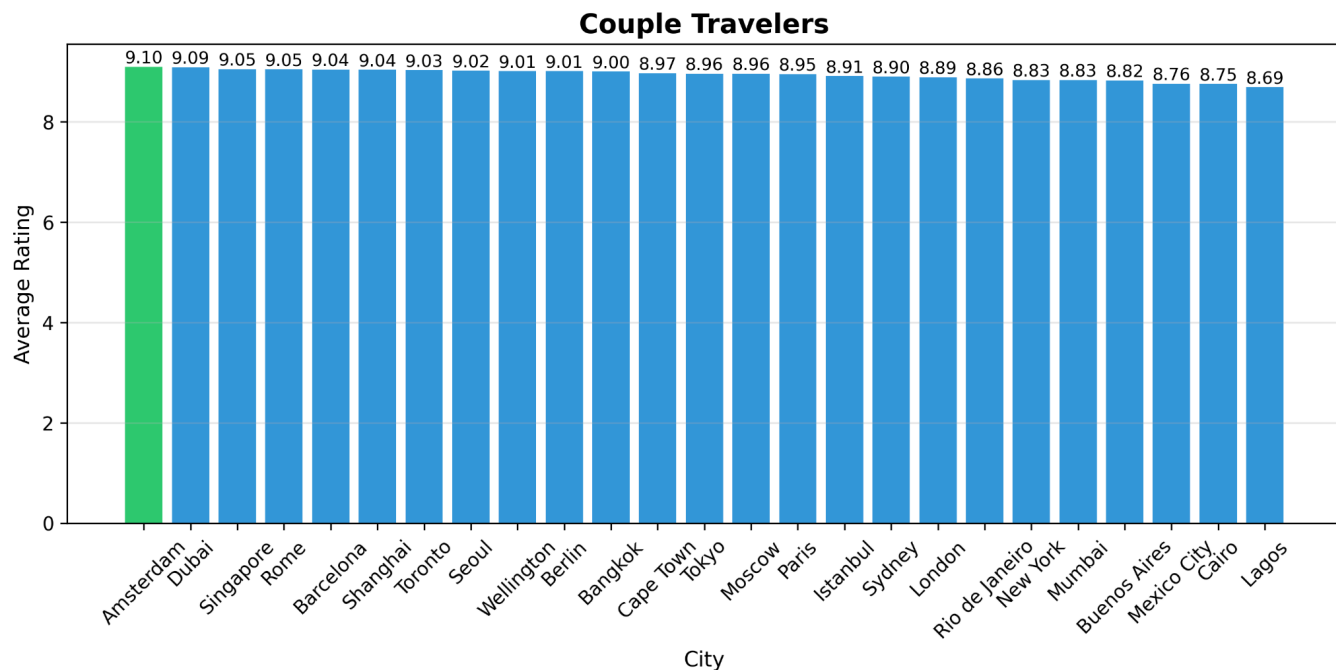
Solo Travelers → Amsterdam (9.108) - Second Highest

- Value independence, safety, ease of navigation, social opportunities
 - Amsterdam's compact size, walkability, English proficiency, and social atmosphere appeal to solo travelers
 - Score suggests excellent solo travel infrastructure
-

Visualization:

Plotted is a sub-graph for each traveller type showing the sorted scores for each city, with the best city highlighted.





Question 2

What are the top 3 countries with the best value-for-money score per traveler's age group?

Discussion:

To answer this question, we need to access the age group from the users table, the country from the hotels table, and we want the value for money score which is found in the reviews table; hence, we use the merged dataframe of the 3 tables using the user_id and hotel_id to merge. Then we want to group by age group and country so that we can have all '18-24' together for example and within the '18-24' we have all the countries grouped together and for each country we get all its value for money scores together. Next we find the mean as there might be several users belonging to the same age group who visited the same country, so we want to get the average of their scores. After that we sort with the mean score descendingly so we get the highest score first and age group ascendingly to get the ages in order. Next we group by the age group and select the first 3 to have each age group together with the top 3 countries for it beside it.

Result:

AGE GROUP	COUNTRY	VALUE FOR MONEY SCORE
18-24	CHINA	8.706926
18-24	NETHERLANDS	8.704911
18-24	CANADA	8.661261
25-34	CHINA	8.727941
25-34	NETHERLANDS	8.683442
25-34	SPAIN	8.625074
35-44	CHINA	8.700799
35-44	NETHERLANDS	8.693268
35-44	NEW ZEALAND	8.646429
45-54	CHINA	8.722112

45-54	NEW ZEALAND	8.674194
45-54	NETHERLANDS	8.647458
55+	NETHERLANDS	8.698537
55+	NEW ZEALAND	8.629384
55+	CHINA	8.601000

Analysis:

1. Overall Patterns Across All Age Groups

Dominant Countries:

- **China:** Appears in all 5 age groups (100% presence)
- **Netherlands:** Appears in all 5 age groups (100% presence)
- **New Zealand:** Appears in 3 out of 5 age groups (60% presence)
- **Canada, Spain:** Each appear in only 1 age group (20% presence)

Key Insight: China and Netherlands consistently deliver strong value-for-money across all demographics, while New Zealand excels for older travelers.

2. Age Group Breakdown

18-24 (Young Adults)

Rank	Country	Score	Analysis
1	China	8.707	Highest score for this age group
2	Netherlands	8.705	Very close to China (0.002 difference)
3	Canada	8.661	Lowest of top 3, but still competitive

Insights:

- Extremely tight competition between China and Netherlands (only 0.002 points apart)

- Young travelers find similar value in Asia (China) and Europe (Netherlands)
 - Canada ranks 3rd but is 0.04-0.05 points behind the leaders
 - Budget-conscious demographic: Young travelers prioritize affordability, and these countries deliver
-

25-34 (Young Professionals)

Rank	Country	Score	Analysis
1	China	8.728	Highest score across all groups
2	Netherlands	8.683	Solid second place
3	Spain	8.625	Only appearance of Spain

Insights:

- China peaks here with its highest score (8.728) - young professionals find exceptional value
 - Gap between China and Netherlands widens to 0.045 points
 - Spain appears only for 25-34 age group - suggests it appeals specifically to young professionals
 - This age group gives the highest overall scores, indicating satisfaction with value
-

35-44 (Established Professionals/Young Families)

Rank	Country	Score	Analysis
1	China	8.701	Continues strong performance
2	Netherlands	8.693	Consistent presence
3	New Zealand	8.646	First appearance of New Zealand

Insights:

- China and Netherlands remain dominant (difference: 0.008)

- New Zealand enters the rankings - appeals to mid-career travelers and families
- Scores slightly lower than 25-34 group - this demographic may be more critical or have different expectations
- Gap between 1st and 3rd place widens to 0.055 points

45-54 (Mid-Career/Pre-Retirement)

Rank	Country	Score	Analysis
1	China	8.722	Second highest score overall
2	New Zealand	8.674	Strong showing
3	Netherlands	8.647	Drops to 3rd for first time

Insights:

- Netherlands drops to 3rd place - first time it's not in top 2
- New Zealand rises to 2nd - increasingly appealing to older travelers
- China maintains dominance with 0.048-point lead over New Zealand
- This shift suggests preference changes with age: mid-career travelers value New Zealand's offerings more

55+ (Retirees/Seniors)

Rank	Country	Score	Analysis
1	Netherlands	8.699	Returns to top spot
2	New Zealand	8.629	Solid second
3	China	8.601	Lowest score across all groups

Insights:

- Major shift: Netherlands reclaims #1, China drops to 3rd
- China's weakest performance (8.601) - suggests seniors value different factors

- New Zealand remains strong (2nd place)
 - Widest spread in this group (0.098 between 1st and 3rd) - seniors have more varied preferences
 - Pattern suggests seniors prioritize comfort, accessibility, familiarity over pure affordability
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3. Country-Specific Analysis

China (Appears in all 5 groups)

Age Group	Rank	Score	Trend
18-24	1st	8.707	Strong start
25-34	1st	8.728	Peak
35-44	1st	8.701	Slight decline
45-54	1st	8.722	Recovery
55+	3rd	8.601	Significant drop

Insights:

- Dominates younger demographics (18-54)
 - Peak performance with 25-34 age group (8.728)
 - Notable decline for seniors (55+) - drops to 3rd place with lowest score (8.601)
 - Why seniors rate it lower: Possible language barriers, cultural differences, travel comfort concerns
 - Overall: Excellent value for younger, more adventurous travelers
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Netherlands (Appears in all 5 groups)

Age Group	Rank	Score	Trend
18-24	2nd	8.705	Strong

25-34	2nd	8.683	Consistent
35-44	2nd	8.693	Stable
45-54	3rd	8.647	Dip
55+	1st	8.699	Recovery to top

Insights:

- Most consistent performer - always in top 3
 - Preferred by seniors (8.699) - reclaims top spot for 55+
 - Brief dip for 45-54 age group (drops to 3rd)
 - Why seniors prefer it: Familiarity, English proficiency, excellent infrastructure, accessibility
 - Overall: Reliable choice across all ages, especially older travelers
-

New Zealand (Appears in 3 groups: 35-44, 45-54, 55+)

Age Group	Rank	Score	Trend
35-44	3rd	8.646	First appearance
45-54	2nd	8.674	Improvement
55+	2nd	8.629	Consistent

Insights:

- Appeals exclusively to older travelers (35+)
 - Absent from younger age groups (18-34) - suggests it's not seen as budget-friendly for youth
 - Improves from 35-44 to 45-54 (rank rises from 3rd to 2nd)
 - Why older travelers prefer it: Scenic beauty, safety, English-speaking, suitable for mature/retirement travel
 - Overall: Premium value destination for older demographics
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Canada (Appears once: 18-24)

- Only appears for 18-24 age group (3rd place, 8.661)
 - Why: Likely appeals to young backpackers, working holiday visa holders, student travelers
 - Notably absent from other age groups - suggests older travelers find better value elsewhere
-

Spain (Appears once: 25-34)

- Only appears for 25-34 age group (3rd place, 8.625)
 - Why: Appeals to young professionals seeking culture, nightlife, beaches, affordability
 - Notably absent from other age groups - narrow appeal to specific demographic
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4. Score Distribution Analysis

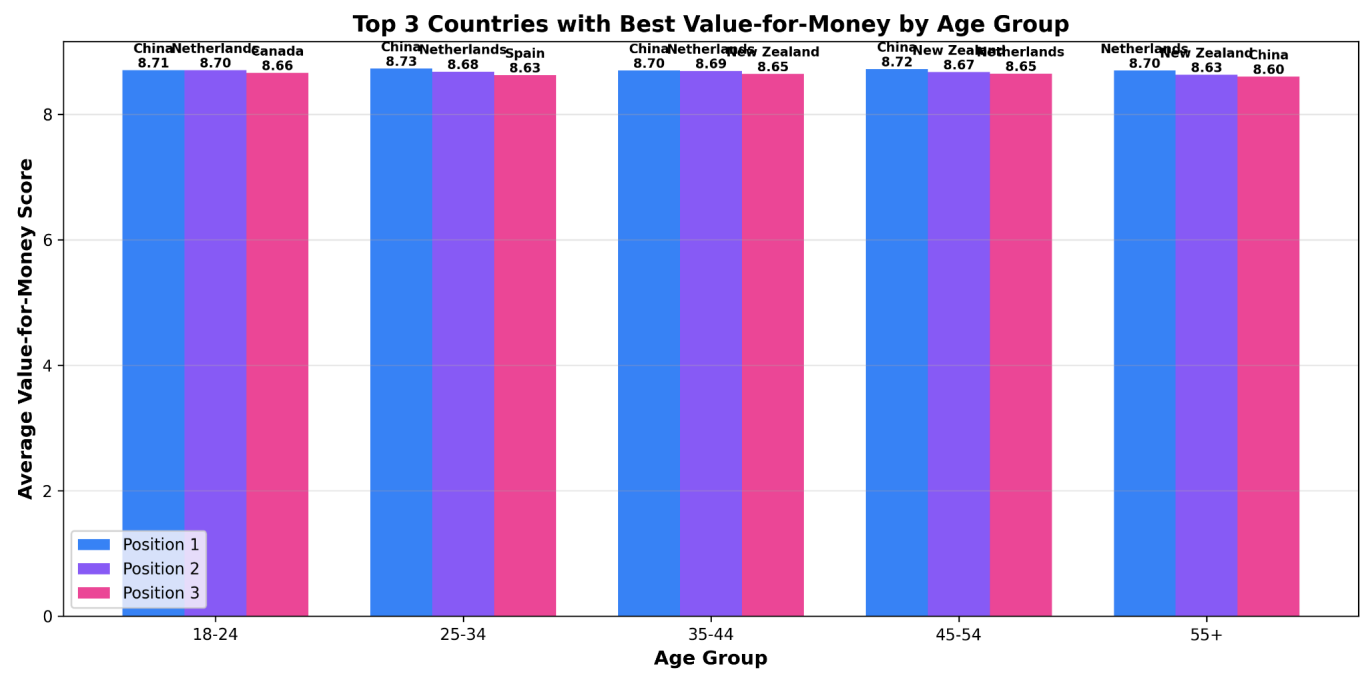
Overall Score Range:

- **Highest: China for 25-34 (8.728)**
- **Lowest: China for 55+ (8.601)**
- **Total Range: 0.127 points**

Observations:

- Very narrow range (0.127) - all top countries provide excellent value
 - Scores cluster around 8.6-8.7 - high satisfaction overall
 - Tightest competition: 18-24 age group (only 0.046 spread)
 - Widest gap: 55+ age group (0.098 spread) - seniors have more divergent preferences
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Visualization:



Model Features

Version 1

Features Used:

Raw:

```
score_overall
score_cleanliness
score_comfort
score_facilities
score_location
score_staff
score_value_for_money
cleanliness_base
comfort_base
facilities_base
location_base
staff_base
value_for_money_base
```

Encoded:

```
user_gender_Female
user_gender_Male
user_gender_Other
age_group_18-24
age_group_25-34
age_group_35-44
age_group_45-54
age_group_55+'
traveller_Business
traveller_Couple
traveller_Family
traveller_Solo
```

Engineered:

```
star_rating_normalized = = star_rating * 2 (to convert to 1-10 scale)
cleanliness_diff = score_cleanliness - cleanliness_base
comfort_diff = score_comfort - comfort_base
facilities_diff = score_facilities - facilities_base
```

```
location_diff = score_location - location_base
staff_diff = score_staff - staff_base
value_diff = score_value_for_money - value_for_money_base
overall_diff = score_overall - star_rating_normalized
```

Model Used:

Model:

LogisticRegression

Parameters:

```
multi_class='multinomial',
solver='lbfgs',
max_iter=1000,
random_state=42
```

Reported Metrics:

Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1-Score: 1.0

Classification Report:

	precision	recall	f1-score	support
Africa	1.00	1.00	1.00	1226
East_Asia	1.00	1.00	1.00	1216
Eastern_Europe	1.00	1.00	1.00	394
Middle_East	1.00	1.00	1.00	797
North_America	1.00	1.00	1.00	792
North_America_Mexico	1.00	1.00	1.00	401
Oceania	1.00	1.00	1.00	803
South_America	1.00	1.00	1.00	784
South_Asia	1.00	1.00	1.00	398
Southeast_Asia	1.00	1.00	1.00	814
Western_Europe	1.00	1.00	1.00	2375
accuracy			1.00	10000
macro avg	1.00	1.00	1.00	10000
weighted avg	1.00	1.00	1.00	10000

Comments:

The metrics all being 100% indicates that the model is memorizing and not learning. This suggests that the used features are doing a data leakage and so we investigated more combinations of different features in the following versions.

Version 2

Features Used:

Raw:

```
score_overall  
score_cleanliness  
score_comfort  
score_facilities  
score_location  
score_staff  
score_value_for_money
```

Encoded:

```
user_gender_Female  
user_gender_Male  
user_gender_Other  
age_group_18-24  
age_group_25-34  
age_group_35-44  
age_group_45-54  
age_group_55+'  
traveller_Business  
traveller_Couple  
traveller_Family  
traveller_Solo
```

Model Used:

Model:

LogisticRegression

Parameters:

```
multi_class='multinomial',  
solver='lbfgs',  
max_iter=1000,  
random_state=42
```

Reported Metrics:

Accuracy: 0.3375
Precision: 0.28316603326006934
Recall: 0.3375
F1-Score: 0.2715156746303661

Classification Report:

	precision	recall	f1-score	support
Africa	0.30	0.49	0.37	1226
East_Asia	0.31	0.43	0.36	1216
Eastern_Europe	0.20	0.02	0.04	394
Middle_East	0.30	0.05	0.09	797
North_America	0.22	0.10	0.14	792
North_America_Mexico	0.30	0.06	0.10	401
Oceania	0.00	0.00	0.00	803
South_America	0.31	0.20	0.24	784
South_Asia	0.00	0.00	0.00	398
Southeast_Asia	0.43	0.19	0.26	814
Western_Europe	0.37	0.75	0.50	2375
accuracy			0.34	10000
macro avg	0.25	0.21	0.19	10000
weighted avg	0.28	0.34	0.27	10000

Comments:

We guessed that the hotel-based features are the ones causing the data leakage so we tried removing them as well as the engineered features, since they are the differences between the hotel base scores and the given reviews. As a result, the accuracy drastically decreased to 33.8%, we also observed a significant decrease in the precision, recall and f1 score all becoming in the range 27-33%. So we were not satisfied with these results as well and investigated more options.

Version 3

Features Used:

Raw:

```
score_overall
score_cleanliness
score_comfort
score_facilities
score_location
score_staff
score_value_for_money
```

Encoded:

```
user_gender_Female
user_gender_Male
user_gender_Other
age_group_18-24
age_group_25-34
age_group_35-44
age_group_45-54
age_group_55+'
traveller_Business
traveller_Couple
traveller_Family
traveller_Solo
```

Engineered:

```
cleanliness_diff = score_cleanliness - cleanliness_base
comfort_diff = score_comfort - comfort_base
facilities_diff = score_facilities - facilities_base
location_diff = score_location - location_base
staff_diff = score_staff - staff_base
value_diff = score_value_for_money - value_for_money_base
overall_diff = score_overall - star_rating_normalized
```

Where `star_rating_normalized = star_rating * 2` (to convert to 1-10 scale)

Model Used:

Model:

`LogisticRegression`

Parameters:

```
multi_class='multinomial',  
solver='lbfgs',  
max_iter=1000,  
random_state=42
```

Reported Metrics:

Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1-Score: 1.0

Classification Report:

	precision	recall	f1-score	support
Africa	1.00	1.00	1.00	1226
East_Asia	1.00	1.00	1.00	1216
Eastern_Europe	1.00	1.00	1.00	394
Middle_East	1.00	1.00	1.00	797
North_America	1.00	1.00	1.00	792
North_America_Mexico	1.00	1.00	1.00	401
Oceania	1.00	1.00	1.00	803
South_America	1.00	1.00	1.00	784
South_Asia	1.00	1.00	1.00	398
Southeast_Asia	1.00	1.00	1.00	814
Western_Europe	1.00	1.00	1.00	2375
accuracy			1.00	10000
macro avg	1.00	1.00	1.00	10000
weighted avg	1.00	1.00	1.00	10000

Comments:

For this version, we tried to find a middle ground between versions 1 and 2. We thought maybe the base hotel scores are causing the issue, but the differences between them and the actual reviews could provide deeper insights and not be totally unique so as not to give data leakage. However, we were proved wrong when the metrics were 100% once again. It looks like more investigation is needed.

Version 4

Features Used:

Raw:

```
score_overall
score_cleanliness
score_comfort
score_facilities
score_location
score_staff
score_value_for_money
cleanliness_base
comfort_base
facilities_base
location_base
staff_base
value_for_money_base
```

Encoded:

```
user_gender_Female
user_gender_Male
user_gender_Other
age_group_18-24
age_group_25-34
age_group_35-44
age_group_45-54
age_group_55+'
traveller_Business
traveller_Couple
traveller_Family
traveller_Solo
```

Engineered:

```
star_rating_normalized = star_rating * 2 (to convert to 1-10 scale)
```

Model Used:

Model:

`LogisticRegression`

Parameters:

```
multi_class='multinomial',  
solver='lbfgs',  
max_iter=1000,  
random_state=42
```

Reported Metrics:

Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1-Score: 1.0

Classification Report:

	precision	recall	f1-score	support
Africa	1.00	1.00	1.00	1226
East_Asia	1.00	1.00	1.00	1216
Eastern_Europe	1.00	1.00	1.00	394
Middle_East	1.00	1.00	1.00	797
North_America	1.00	1.00	1.00	792
North_America_Mexico	1.00	1.00	1.00	401
Oceania	1.00	1.00	1.00	803
South_America	1.00	1.00	1.00	784
South_Asia	1.00	1.00	1.00	398
Southeast_Asia	1.00	1.00	1.00	814
Western_Europe	1.00	1.00	1.00	2375
accuracy			1.00	10000
macro avg	1.00	1.00	1.00	10000
weighted avg	1.00	1.00	1.00	10000

Comments:

For this version, we wanted to prove our hypothesis that the hotel base scores are the ones causing the data leakage so we added them and removed the differences and the metrics were still 100% so here we identified a major issue. In the next versions we tried to explore more feature engineering and more reasoning.

Version 5

Features Used:

Raw:

```
score_overall
score_cleanliness
score_comfort
score_facilities
score_location
score_staff
score_value_for_money
```

Encoded:

```
user_gender_Female
user_gender_Male
user_gender_Other
age_group_18-24
age_group_25-34
age_group_35-44
age_group_45-54
age_group_55+'
traveller_Business
traveller_Couple
traveller_Family
traveller_Solo
```

Engineered:

```
avg_service_score = (score_staff + score_cleanliness) / 2
avg_facility_score = (score_facilities + score_comfort) / 2

quality_per_dollar = (score_cleanliness + score_comfort + score_facilities)
                    / (3 * score_value_for_money + 0.01)

location_premium = score_location - score_value_for_money

quality_consistency = [score_cleanliness, score_comfort, score_facilities,
score_staff].std(axis=1)
```

```

service_facility_ratio = (
    (score_staff + score_cleanliness) /
    (score_facilities + score_comfort + 0.01)
)
experience_score = (score_location + score_staff) / 2
physical_score = (score_cleanliness + score_facilities) / 2
value_satisfaction = score_overall - score_value_for_money

```

Model Used:

Model:

LogisticRegression

Parameters:

```

multi_class='multinomial',
solver='lbfgs',
max_iter=1000,
random_state=42

```

Reported Metrics:

Accuracy: 0.3382

Precision: 0.2837412051148293

Recall: 0.3382

F1-Score: 0.27551142979077414

Classification Report:

	precision	recall	f1-score	support
Africa	0.30	0.49	0.37	1226
East_Asia	0.32	0.42	0.36	1216
Eastern_Europe	0.15	0.02	0.03	394
Middle_East	0.33	0.10	0.15	797
North_America	0.21	0.09	0.13	792
North_America_Mexico	0.33	0.06	0.11	401
Oceania	0.00	0.00	0.00	803
South_America	0.32	0.21	0.25	784
South_Asia	0.00	0.00	0.00	398
Southeast_Asia	0.42	0.19	0.26	814
Western_Europe	0.37	0.75	0.50	2375
accuracy			0.34	10000
macro avg	0.25	0.21	0.20	10000
weighted avg	0.28	0.34	0.28	10000

Comments:

After detecting that the hotel base scores are the ones responsible for the data leakage, we decided to eliminate them completely, however when we did that in version 2 and depended solely on the review scores, the metrics reported were so bad. Naturally, the first course of action that came to our mind was to engineer some features using the review scores to help the model identify some patterns that it may not detect trivially. To our surprise, the metrics did not improve at all from version 2. This started to make us question the model, maybe logistic regression is too simple?

Version 6

Features Used:

Raw:

```
score_overall
score_cleanliness
score_comfort
score_facilities
score_location
score_staff
score_value_for_money
```

Encoded:

```
user_gender_Female
user_gender_Male
user_gender_Other
age_group_18-24
age_group_25-34
age_group_35-44
age_group_45-54
age_group_55+'
traveller_Business
traveller_Couple
traveller_Family
traveller_Solo
```

Engineered:

```
avg_service_score = (score_staff + score_cleanliness) / 2
```

```
avg_facility_score = (score_facilities + score_comfort) / 2

quality_per_dollar = (score_cleanliness + score_comfort + score_facilities)
                    / (3 * score_value_for_money + 0.01)

location_premium = score_location - score_value_for_money

quality_consistency = [score_cleanliness, score_comfort, score_facilities,
score_staff].std(axis=1)

service_facility_ratio = (
    (score_staff + score_cleanliness) /
    (score_facilities + score_comfort + 0.01)
)

experience_score = (score_location + score_staff) / 2
physical_score = (score_cleanliness + score_facilities) / 2
value_satisfaction = score_overall - score_value_for_money
```

Model Used:

Model:

RandomForestClassifier

Parameters:

```
n_estimators=200,
max_depth=20,
min_samples_split=10,
min_samples_leaf=5,
class_weight='balanced',
random_state=42,
n_jobs=-1,
verbose=1
```

Reported Metrics:

Accuracy: 0.3323
Precision: 0.36121307027076954
Recall: 0.3323
F1-Score: 0.3402560144542057

Classification Report:

	precision	recall	f1-score	support
Africa	0.43	0.29	0.34	1226
East_Asia	0.44	0.38	0.40	1216
Eastern_Europe	0.13	0.24	0.17	394
Middle_East	0.36	0.31	0.33	797
North_America	0.34	0.34	0.34	792
North_America_Mexico	0.19	0.35	0.25	401
Oceania	0.26	0.17	0.20	803
South_America	0.25	0.32	0.28	784
South_Asia	0.10	0.13	0.11	398
Southeast_Asia	0.27	0.37	0.31	814
Western_Europe	0.51	0.43	0.47	2375
accuracy			0.33	10000
macro avg	0.30	0.30	0.29	10000
weighted avg	0.36	0.33	0.34	10000

Comments:

For version 6, we decided to change the model completely. We figured that logistic regression may be too simple to detect complex patterns in the features. So that we don't get too complex, we limited ourselves to models covered in the lab, so our options were neural networks or random forest. We decided to go with random forest since neural networks have so many tunable parameters including the architecture itself (the amount of layers, neurons per layer..etc) so this made random forest seem simpler and more effective. The results showed a very slight increase in precision and f1 score, so this led us to 2 conclusions; the first is that random forest performs slightly better than logistic regression, and the second is that our engineered features do not add any extra information and so we should remove them.

Version 7

Features Used:

Raw:

facilities_base

Encoded:

user_gender_Female

user_gender_Male

user_gender_Other

age_group_18-24

age_group_25-34

age_group_35-44

age_group_45-54

age_group_55+'

traveller_Business

traveller_Couple

traveller_Family

traveller_Solo

Model Used:

Model:

RandomForestClassifier

Parameters:

```
n_estimators=200,  
max_depth=20,  
min_samples_split=10,  
min_samples_leaf=5,  
class_weight='balanced',  
random_state=42,  
n_jobs=-1,  
verbose=1
```

Reported Metrics:

Accuracy: 0.4416
Precision: 0.5567706255276289
Recall: 0.4416
F1-Score: 0.3505319628238868

Classification Report:

	precision	recall	f1-score	support
Africa	0.41	0.02	0.04	1226
East_Asia	0.92	0.34	0.50	1216
Eastern_Europe	0.33	0.99	0.49	394
Middle_East	0.51	0.99	0.67	797
North_America	0.41	0.94	0.57	792
North_America_Mexico	0.34	1.00	0.50	401
Oceania	0.33	0.28	0.30	803
South_America	0.48	0.48	0.48	784
South_Asia	0.34	0.99	0.51	398
Southeast_Asia	0.59	0.73	0.65	814
Western_Europe	0.71	0.03	0.05	2375
accuracy			0.44	10000
macro avg	0.49	0.62	0.43	10000
weighted avg	0.56	0.44	0.35	10000

Comments:

After we decided to get rid of the extra engineered features, we thought about revisiting the hotel base scores but in a different way. The driving question was “what exactly about these values causes the data leakage?” For a small test we used only one of them, namely, the facilities base, and to our surprise we got an accuracy of 44%, which is a notable improvement from the previous versions. To make sure that this is not a coincidence, we ran another test.

Version 8

Features Used:

Raw:

location_base

Encoded:

user_gender_Female

user_gender_Male

user_gender_Other

age_group_18-24

age_group_25-34

age_group_35-44

age_group_45-54

age_group_55+'

traveller_Business

traveller_Couple

traveller_Family

traveller_Solo

Model Used:

Model:

RandomForestClassifier

Parameters:

```
n_estimators=200,  
max_depth=20,  
min_samples_split=10,  
min_samples_leaf=5,  
class_weight='balanced',  
random_state=42,  
n_jobs=-1,  
verbose=1
```

Reported Metrics:

Accuracy: 0.4929
Precision: 0.5326835741264364
Recall: 0.4929
F1-Score: 0.45910425012135586

Classification Report:

	precision	recall	f1-score	support
Africa	0.48	0.34	0.40	1226
East_Asia	0.58	0.49	0.53	1216
Eastern_Europe	0.25	0.47	0.33	394
Middle_East	0.49	0.94	0.64	797
North_America	0.57	0.79	0.66	792
North_America_Mexico	0.26	0.55	0.35	401
Oceania	0.48	0.46	0.47	803
South_America	0.34	0.22	0.26	784
South_Asia	1.00	1.00	1.00	398
Southeast_Asia	0.50	0.99	0.67	814
Western_Europe	0.64	0.17	0.26	2375
accuracy			0.49	10000
macro avg	0.51	0.58	0.51	10000
weighted avg	0.53	0.49	0.46	10000

Comments:

Version 8 is just version 7 however we used a different hotel base metric, namely, location base. We got an accuracy of 49% and this made us take a closer look at the data. Our conclusion was that the 6 hotel base scores together form a unique permutation per hotel which makes the model memorize and not learn, but using single scores does not provide the same uniqueness aspect. So this led us to think of a more robust solution, which is to combine the scores in a way that doesn't introduce uniqueness.

Version 9

Features Used:

Encoded:

```
user_gender_Female
user_gender_Male
user_gender_Other
age_group_18-24
age_group_25-34
age_group_35-44
age_group_45-54
age_group_55+'
traveller_Business
traveller_Couple
traveller_Family
traveller_Solo
```

Engineered:

```
avg_hotel_score = [cleanliness_base, comfort_base, facilities_base,
location_base, staff_base, value_for_money_base].mean(axis=1)
```

Model Used:

Model:

```
RandomForestClassifier
```

Parameters:

```
    n_estimators=200,
    max_depth=20,
    min_samples_split=10,
    min_samples_leaf=5,
    class_weight='balanced',
    random_state=42,
    n_jobs=-1,
    verbose=1
```


Reported Metrics:

Accuracy: 0.5968
Precision: 0.7191609694067974
Recall: 0.5968
F1-Score: 0.5514820903201074

Classification Report:

	precision	recall	f1-score	support
Africa	1.00	0.34	0.51	1226
East_Asia	0.48	0.03	0.06	1216
Eastern_Europe	0.33	0.98	0.50	394
Middle_East	0.65	0.99	0.79	797
North_America	0.38	0.77	0.51	792
North_America_Mexico	0.50	1.00	0.66	401
Oceania	0.64	0.97	0.77	803
South_America	0.77	0.72	0.74	784
South_Asia	1.00	1.00	1.00	398
Southeast_Asia	0.51	0.97	0.67	814
Western_Europe	0.97	0.33	0.49	2375
accuracy			0.60	10000
macro avg	0.66	0.74	0.61	10000
weighted avg	0.72	0.60	0.55	10000

Comments:

After some observation of the data, we created a new feature which is the average of all the hotel base scores. These averages are not unique and so they do not act as identifiers for the hotels and can be used to train the model without causing data leakage. The result was an accuracy of almost 60% and precision of almost 72%, so it is a notable improvement.

Version 10

Features Used:

Raw:

```
score_overall  
score_cleanliness  
score_comfort  
score_facilities  
score_location  
score_staff  
score_value_for_money
```

Encoded:

```
user_gender_Female  
user_gender_Male  
user_gender_Other  
age_group_18-24  
age_group_25-34  
age_group_35-44  
age_group_45-54  
age_group_55+'  
traveller_Business  
traveller_Couple  
traveller_Family  
traveller_Solo
```

Engineered:

```
avg_hotel_score = [cleanliness_base, comfort_base, facilities_base,  
location_base, staff_base, value_for_money_base].mean(axis=1)
```

Model Used:

Model:

```
RandomForestClassifier
```

Parameters:

```
n_estimators=200,
```

```

max_depth=20,
min_samples_split=10,
min_samples_leaf=5,
class_weight='balanced',
random_state=42,
n_jobs=-1,
verbose=1

```

Reported Metrics:

Accuracy: 0.8221
 Precision: 0.8525417882229543
 Recall: 0.8221
 F1-Score: 0.8234069233249227

Classification Report:

	precision	recall	f1-score	support
Africa	0.97	0.59	0.73	1226
East_Asia	0.79	0.66	0.72	1216
Eastern_Europe	0.58	0.93	0.72	394
Middle_East	0.92	0.99	0.95	797
North_America	0.85	0.88	0.86	792
North_America_Mexico	0.57	0.96	0.72	401
Oceania	0.79	0.82	0.80	803
South_America	0.93	0.96	0.94	784
South_Asia	1.00	1.00	1.00	398
Southeast_Asia	0.62	0.91	0.74	814
Western_Europe	0.95	0.80	0.87	2375
accuracy			0.82	10000
macro avg	0.81	0.86	0.82	10000
weighted avg	0.85	0.82	0.82	10000

Comments:

The result for version 9 was pretty good, however this was only taking into account the hotel base scores and not the review scores, so after adding both, we got an accuracy of 82.2% which is the highest reported yet without causing data leakage.